

Gaussian centroid partitions with arbitrary prescribed masses: a universal two-times-collision-probability approximation

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Abstract

For any positive probability vector $p = (p_1, \dots, p_k)$ and $d \geq k - 1$, consider the maximum

$$\mathcal{M}_d(p) = \sup_{\substack{\mathbb{R}^d = \bigsqcup_{i=1}^k A_i \\ \gamma_d(A_i) = p_i}} \sum_{i=1}^k \left\| \int_{A_i} x d\gamma_d(x) \right\|^2.$$

Set $t_i = \Phi^{-1}(1 - p_i)$, $\varphi(t) = (2\pi)^{-1/2}e^{-t^2/2}$, $U(p) = \sum_i \varphi(t_i)^2$, and $Q(p) = \sum_i p_i^2$.

Main theorem. For *every* positive mass vector,

$$U(p) - 2Q(p) \leq \mathcal{M}_d(p) \leq U(p) \quad (d \geq k - 1). \quad (1)$$

The lower bound comes from an explicit disjoint, axis-aligned, sequential-threshold partition; it does *not* assume any Gaussian simplex conjecture. The gap of at most $2Q(p)$ is dimension- and mass-distribution-independent.

We also establish the fully explicit mass-entropy approximation

$$B(p) - 6Q(p) \leq \mathcal{M}_d(p) \leq B(p) + 3Q(p), \quad B(p) = \sum_i p_i^2 \left[2 \log \frac{1}{p_i} - \log^+ \log \frac{1}{p_i} \right], \quad (2)$$

where $\log^+ u = \max(0, \log u)$ for $u > 0$. This determines the leading two logarithmic terms of the unrestricted Gaussian first-moment optimum for **all** small, even highly nonuniform prescribed masses, with an absolute remainder proportional to $Q(p)$.

The input problem and classical one-cell Gaussian halfspace bound have historical antecedents. Originality and publication priority are not asserted prior to a full literature review.

1. Statement and exact construction

Let $k \geq 2$, $p_i > 0$, $\sum_i p_i = 1$. We optimize over measurable ordered partitions modulo γ_d -null sets. No convexity, conical restriction, regularity, symmetry, or equality of cell masses is assumed. Write $Z \sim N(0, 1)$,

$$\varphi(t) = (2\pi)^{-1/2}e^{-t^2/2}, \quad \bar{\Phi}(t) = \Pr(Z \geq t), \quad t_q = \bar{\Phi}^{-1}(q), \quad h(q) = \frac{\varphi(t_q)}{q} \quad (0 < q < 1). \quad (3)$$

Thus $h(q) = \mathbb{E}[Z \mid Z \geq t_q]$, and $U(p) = \sum_i p_i^2 h(p_i)^2$.

Relabel the prescribed masses so $p_1 \geq p_2 \geq \dots \geq p_k > 0$, and put

$$S_i = \sum_{j=i}^k p_j, \quad S_1 = 1, \quad q_i = \frac{p_i}{S_i} \in (0, 1) \quad (1 \leq i < k). \quad (4)$$

In $\mathbb{R}^{k-1}$, with standard independent Gaussian coordinates $Z_1, \dots, Z_{k-1}$, define the staircase cells

$$\begin{aligned} A_i &= \bigcap_{j < i} \{Z_j < t_{q_j}\} \cap \{Z_i \geq t_{q_i}\} \quad (1 \leq i < k), \\ A_k &= \bigcap_{j < k} \{Z_j < t_{q_j}\}. \end{aligned} \quad (5)$$

These cells are disjoint and exhaust the space, up to a union of Gaussian-null coordinate hyperplanes.

Lemma 1 (exact masses and diagonal moments). The cells (5) satisfy, identically,

$$\gamma_{k-1}(A_i) = p_i \quad (1 \leq i \leq k), \quad \int_{A_i} x_i d\gamma_{k-1}(x) = p_i h(q_i) \quad (1 \leq i < k). \quad (6)$$

Consequently, their objective $\mathcal{P}_{\text{stair}}(p)$ obeys

$$\mathcal{P}_{\text{stair}}(p) \geq \sum_{i=1}^{k-1} p_i^2 h(q_i)^2. \quad (7)$$

Proof. Since $1 - q_i = (S_i - p_i)/S_i = S_{i+1}/S_i$, independence yields

$$\Pr(Z_j < t_{q_j} \quad \forall j < i) = \prod_{j < i} (1 - q_j) = S_i.$$

Multiplying by q_i gives $\gamma(A_i) = S_i q_i = p_i$; the last cell has mass $S_k = p_k$. For $i < k$, coordinate Z_i is independent of all earlier threshold constraints, and $\mathbb{E}[Z_i \mathbf{1}_{\{Z_i \geq t\}}] = \varphi(t)$ by one integration by parts. Its contribution to the first moment of A_i is $S_i \varphi(t_{q_i}) = p_i h(q_i)$. The squared Euclidean norm dominates the square of any specified component, proving (7). $\square$

Supplement to Lemma 1 (exact staircase objective). The full first moments of (5) can also be expressed without any multivariate Gaussian integration. Writing $b_{i,j} = \int_{A_i} x_j d\gamma_{k-1}$ for $1 \leq j < k$, one has

$$b_{i,j} = \begin{cases} 0, & i < j, \\ p_j h(q_j), & i = j, \\ -p_i \frac{\varphi(t_{q_j})}{1 - q_j}, & i > j. \end{cases}$$

Indeed, for $i > j$, conditioning on $Z_j < t_{q_j}$ gives $\mathbb{E}[Z_j \mathbf{1}_{\{Z_j < t\}}] = -\varphi(t)$ and the other independent constraints have probability $p_i/(1 - q_j)$. The exact objective is therefore

$$\mathcal{P}_{\text{stair}}(p) = \sum_{j=1}^{k-1} \left[p_j^2 h(q_j)^2 + \frac{\varphi(t_{q_j})^2}{(1 - q_j)^2} \sum_{i=j+1}^k p_i^2 \right]. \quad (7a)$$

The second, nonnegative term was deliberately omitted from the universal envelope proof; it can strengthen instance-specific lower certificates.

2. Two analytic lemmas

The first lemma is an exact, global Lipschitz property of the Gaussian upper-tail hazard. It is the principal analytic observation enabling a constant error bound.

Lemma 2 (2-Lipschitz squared hazard in log tail mass). For every $0 < q_1 \leq q_2 < 1$,

$$0 \leq h(q_1)^2 - h(q_2)^2 \leq 2 \log(q_2/q_1). \quad (8)$$

Proof. Set $q = \bar{\Phi}(t)$ and $h = \varphi(t)/q$. Direct differentiation gives

$$\frac{dh}{dt} = h(h-t), \quad \frac{dt}{d \log(1/q)} = \frac{1}{h}, \quad \frac{d(h^2)}{d \log(1/q)} = 2h(h-t). \quad (9)$$

The conditional mean of a nondegenerate random variable supported strictly above t exceeds t ; hence $h-t > 0$. The truncated standard-normal second moment satisfies

$$\mathbb{E}[Z^2 \mid Z \geq t] = 1 + th,$$

so its strictly positive variance is

$$\text{Var}(Z \mid Z \geq t) = 1 + th - h^2 = 1 - h(h-t) > 0.$$

Thus $0 < h(h-t) < 1$. Integrating (9) over the log-tail-mass interval proves (8). $\square$

Sharpness of the analytic Lipschitz constant. The coefficient 2 in Lemma 2 cannot be replaced by any smaller constant *in that lemma*. To see this, integrate by parts twice:

$$\bar{\Phi}(t) = \varphi(t) \left(\frac{1}{t} - \frac{1}{t^3} + O(t^{-5}) \right) \quad (t \rightarrow +\infty).$$

The error follows from the exact remainder $3 \int_t^\infty \varphi(x)/x^4 dx$, bounded above by $3\varphi(t)/t^5$ using the upper Mills inequality. Consequently $h(t) = \varphi(t)/\bar{\Phi}(t) = t + 1/t + O(t^{-3})$ and $h(t)(h(t)-t) \rightarrow 1$. Equation (9) therefore gives $d(h^2)/d \log(1/q) \rightarrow 2$ as $q \downarrow 0$. This proves sharpness of the scalar Lipschitz coefficient, **not** sharpness of the global partition-envelope coefficient in Theorem 4.

The second lemma is a distribution-free weighted entropy inequality for a decreasing probability vector.

Lemma 3 (ordered residual-mass entropy). For any $p_1 \geq \dots \geq p_k > 0$, $\sum_i p_i = 1$, and $S_i = \sum_{j=i}^k p_j$,

$$\boxed{\sum_{i=1}^k p_i^2 \log \frac{1}{S_i} \leq \sum_{i=1}^k p_i^2 = Q(p)}. \quad (10)$$

Proof. Put $g_i = \log(1/S_i)$. Since S_i decreases, g_i increases, whereas p_i decreases. The weighted negative-covariance identity is

$$\begin{aligned} & \sum_i p_i^2 g_i - \left(\sum_i p_i^2 \right) \left(\sum_i p_i g_i \right) \\ &= \frac{1}{2} \sum_{i,j} p_i p_j (p_i - p_j) (g_i - g_j) \leq 0. \end{aligned} \quad (11)$$

Next put $S_{k+1} = 0$. Because $-\log s$ decreases with $s \in (0, 1]$,

$$p_i \log \frac{1}{S_i} \leq \int_{S_{i+1}}^{S_i} \log \frac{1}{s} ds.$$

Summing the integrals gives

$$\sum_i p_i g_i \leq \int_0^1 \log \frac{1}{s} ds = 1. \quad (12)$$

Combining (11)–(12) proves (10). $\square$

3. Main theorem: additive two-collision-mass gap

Theorem 4 (universal two-collision-mass centroid envelope). For every $k \geq 2$, $d \geq k-1$, and positive prescribed Gaussian cell masses p_i totaling one,

$$\boxed{0 \leq U(p) - \mathcal{M}_d(p) \leq 2Q(p).} \quad (13)$$

More precisely, the explicit partition (5), extended cylindrically to $\mathbb{R}^d$, attains objective at least $U(p) - 2Q(p)$.

Proof. First, the **individual one-cell upper bound** requires no assumption on cell shape. If $A \subseteq \mathbb{R}^d$ has Gaussian mass q and $b = \int_A x d\gamma_d \neq 0$, let $u = b/\|b\|$ and $H = \{x : \langle u, x \rangle \geq t_q\}$. Then $\gamma_d(H) = q$. The pointwise integrand $(\langle u, x \rangle - t_q)(\mathbf{1}_H - \mathbf{1}_A)$ is nonnegative, and the threshold terms cancel after integration because A, H have the same mass. Therefore

$$\|b\| = \int_A \langle u, x \rangle d\gamma_d \leq \int_H \langle u, x \rangle d\gamma_d = \varphi(t_q) = qh(q). \quad (14)$$

It is also true for $b = 0$. Applying this separately to all cells proves $\mathcal{M}_d(p) \leq U(p)$.

For the lower bound, choose the decreasing order (4). Because $q_i = p_i/S_i \geq p_i$, Lemma 2 gives

$$0 \leq h(p_i)^2 - h(q_i)^2 \leq 2 \log \frac{q_i}{p_i} = 2 \log \frac{1}{S_i}. \quad (15)$$

Multiply by p_i^2 , sum over $i < k$, and apply Lemma 1. We have

$$U(p) - \mathcal{P}_{\text{stair}}(p) \leq 2 \sum_{i < k} p_i^2 \log \frac{1}{S_i} + p_k^2 h(p_k)^2. \quad (16a)$$

The final omitted cell satisfies a stronger bound than ordinary Cauchy–Schwarz. For the upper-tail event $H = \{Z \geq t_q\}$, of mass q , the Gaussian exponential moment identity and conditional Jensen give, for every $\lambda \geq 0$,

$$e^{\lambda^2/2} = \mathbb{E} e^{\lambda Z} \geq q \mathbb{E}[e^{\lambda Z} \mid H] \geq q e^{\lambda h(q)}.$$

Taking logarithms and choosing $\lambda = h(q) > 0$ yields

$$\boxed{\varphi(t_q)^2 = q^2 h(q)^2 \leq 2q^2 \log(1/q).} \quad (16b)$$

Since $S_k = p_k$, the omitted contribution is controlled by the **same residual entropy**. Combining (16a)–(16b) with Lemma 3,

$$\begin{aligned} U(p) - \mathcal{P}_{\text{stair}}(p) &\leq 2 \sum_{i < k} p_i^2 \log \frac{1}{S_i} + 2p_k^2 \log \frac{1}{p_k} \\ &= 2 \sum_{i=1}^k p_i^2 \log \frac{1}{S_i} \leq 2Q(p). \end{aligned} \quad (16)$$

The staircase partition is defined in exactly $k-1$ Gaussian coordinates. Extending it by an independent centered Gaussian factor leaves masses and squared first moments unchanged for all $d \geq k-1$. Taking the supremum over feasible partitions proves (13). $\square$

Interpretation. Each individual Gaussian cell of mass p_i could, in isolation, be a halfspace with squared centroid $\varphi(t_{p_i})^2$. Theorem 4 proves that we can satisfy **all** prescribed masses at once with disjoint cells, losing at most $2 \sum_i p_i^2$ in the sum of these individually optimal squared moments. No numerical search or optimization solver enters.

4. A universal scalar Gaussian quantile estimate

The additive approximation in Theorem 4 combines with a *quantitative* elementary normal-tail estimate. For $0 < q < 1$, define

$$L(q) = \log \frac{1}{q} > 0, \quad \mathfrak{b}(q) = 2L(q) - \max(0, \log L(q)). \quad (17)$$

The positive-part logarithm avoids a spurious $-\log \log(1/q)$ divergence when $q \rightarrow 1$.

Lemma 5 (sharpened universal hazard/entropy comparison). For every $0 < q < 1$,

$$\boxed{\mathfrak{b}(q) - 4 \leq h(q)^2 \leq \mathfrak{b}(q) + 3.} \quad (18)$$

Both constants are absolute and hold without any small-tail asymptotic assumption.

Proof. Set $L = \log(1/q)$, $t = t_q$. By an integration by parts, for every $t > 0$ the familiar Mills bounds are

$$\frac{t}{1+t^2} \varphi(t) \leq \bar{\Phi}(t) \leq \frac{\varphi(t)}{t}. \quad (19)$$

First consider $q \leq 1/10$. Mills' lower bound at $t = 1$ gives $\bar{\Phi}(1) \geq \varphi(1)/2 > 1/10$; indeed $2\pi e < 24 < 25$ using $\pi < 4$ and $e < 3$. Consequently $t > 1$ and $L > 1$. The Mills bounds yield

$$t \leq h(q) \leq t + \frac{1}{t}, \quad \frac{t^2}{2} + \log t + \frac{1}{2} \log(2\pi) \leq L \leq \frac{t^2}{2} + \log(t + 1/t) + \frac{1}{2} \log(2\pi). \quad (20)$$

In particular $t^2 \leq 2L$. Since $t + 1/t \leq 2\sqrt{2L}$, the right inequality gives

$$t^2 \geq 2L - \log L - \log(16\pi). \quad (21)$$

To see $\log(16\pi) < 4$ with strict elementary inequalities, note that $e > 8/3$ and $\pi < 22/7$, while $(8/3)^4 = 4096/81 > 352/7 = 16(22/7)$. Hence $h(q)^2 \geq t^2 > \mathfrak{b}(q) - 4$.

For the upper bound, $h(q)^2 \leq t^2 + 3$. If $t^2 \geq L$, the left inequality in (20) gives $t^2 \leq 2L - \log L - \log(2\pi)$, so $h(q)^2 \leq \mathfrak{b}(q) + 3$. If $t^2 < L$, the elementary inequality $L - \log L \geq 1$ gives $h(q)^2 < L + 3 \leq \mathfrak{b}(q) + 2$. Thus the result holds for all $q \leq 1/10$.

Next suppose $q > 1/10$. Then $L < \log 10$. Conditional Jensen applied to $q e^{\lambda h(q)} \leq \mathbb{E} e^{\lambda Z} = e^{\lambda^2/2}$ at $\lambda = h(q)$ gives $h(q)^2 \leq 2L$. Therefore

$$h(q)^2 - \mathfrak{b}(q) \leq \max(0, \log L) \leq \log \log 10 < 1 < 3.$$

For the lower bound, $h(q)^2 \geq 0$ and we show $\mathfrak{b}(q) < 4$. For $L \leq 1$, $\mathfrak{b}(q) = 2L \leq 2$. For $1 < L < \log 10$, the function $2L - \log L$ is increasing, so its maximum occurs at $L = \log 10$. Now $\log 10 < 7/3$, since $e > 27/10$ (which follows by summing the first five terms of its power series) implies $e^7 > (27/10)^7 > 10^3$. Also $\log 10 > 2$, since $e < 3$, and $\log 2 > 2/3$ by the positive arctanh series. Consequently

$$2 \log 10 - \log \log 10 < \frac{14}{3} - \frac{2}{3} = 4.$$

This proves $\mathfrak{b}(q) - 4 < h(q)^2$ and completes the proof. $\square$

Theorem 6 (unrestricted mass-entropy envelope). For every $k \geq 2$, positive probability vector p , and $d \geq k - 1$, put

$$B(p) = \sum_{i=1}^k p_i^2 \mathfrak{b}(p_i).$$

Then

$$\boxed{B(p) - 6Q(p) \leq \mathcal{M}_d(p) \leq B(p) + 3Q(p).} \quad (22)$$

Proof. Applying Lemma 5 to each mass gives $B(p) - 4Q(p) \leq U(p) \leq B(p) + 3Q(p)$. Theorem 4 gives $U(p) - 2Q(p) \leq \mathcal{M}_d(p) \leq U(p)$. Combining these inequalities proves (22). $\square$

5. Consequences for all small masses

Corollary 7 (two-term entropy expansion without comparable-mass assumptions). Suppose $d \geq k - 1$ and $p_{\max} := \max_i p_i \leq e^{-1}$. Then

$$\mathcal{M}_d(p) = 2 \sum_i p_i^2 \log \frac{1}{p_i} - \sum_i p_i^2 \log \log \frac{1}{p_i} + E(p), \quad |E(p)| \leq 6Q(p). \quad (23)$$

For *any* family of probability vectors with $p_{\max} \rightarrow 0$, regardless of the ratio $p_{\max}/p_{\min}$,

$$\mathcal{M}_d(p) \sim 2 \sum_i p_i^2 \log \frac{1}{p_i}. \quad (24)$$

The logarithmic second term is meaningful: its magnitude divided by $Q(p)$ tends to infinity, whereas the error in (23) is at most $6Q(p)$.

Proof. If $p_i \leq e^{-1}$, then $L(p_i) \geq 1$ and $\log^+ L(p_i) = \log L(p_i)$, so (23) is exactly Theorem 6. Let $L_* = \log(1/p_{\max}) \rightarrow \infty$. For all $L \geq L_*$, $\log L/L$ tends uniformly to zero (because it is decreasing for $L > e$). Also $Q(p)/\sum_i p_i^2 L(p_i) \leq 1/L_*$. Dividing (23) by $2 \sum_i p_i^2 L(p_i)$ proves (24). Finally $\sum_i p_i^2 \log L(p_i) \geq (\log L_*)Q(p)$, proving the assertion about the second term. $\square$

Corollary 8 (near-uniform masses and the exact log-log coefficient). Fix $0 < a \leq 1 \leq b < \infty$. Uniformly for all $k \rightarrow \infty$, vectors with $a/k \leq p_i \leq b/k$, and $d \geq k - 1$,

$$\mathcal{M}_d(p) = (2 \log k - \log \log k - \log(4\pi)) \sum_i p_i^2 + O_{a,b}(1/k). \quad (25)$$

No equality between the masses is assumed.

Proof. Write $p_i = c_i/k$, where $c_i \in [a, b]$. Then $L(p_i) = \log k - \log c_i$, so uniformly in i ,

$$2L(p_i) - \log L(p_i) = 2 \log k - \log \log k + O_{a,b}(1).$$

The constant term $-\log(4\pi)$ is **not identified** by the stated $O_{a,b}(1/k)$ precision; inserting it in (25) merely aligns the expression with the normal quantile convention. Only the coefficients of the leading and log-log terms are determined here. Moreover $Q(p) \leq b/k$, since $\sum_i p_i^2 \leq (\max_i p_i) \sum_i p_i$. Now apply (23). $\square$

Corollary 9 (constructive global relative saturation). If $p_{\max} \leq e^{-10}$, set $L_* = \log(1/p_{\max}) \geq 40$. Then

$$\frac{\mathcal{M}_d(p)}{U(p)} \geq 1 - \frac{2}{2L_* - \log L_* - 4} \geq 1 - \frac{2}{L_*} \quad (d \geq k - 1). \quad (26)$$

The *explicit staircase partition* (5) satisfies the same ratio. In particular, as $p_{\max} \rightarrow 0$, it is asymptotically optimal for the full unconstrained fixed-mass Gaussian partition problem, even for arbitrarily heterogeneous masses.

Proof. By Lemma 5, for $L_i = L(p_i) \geq L_*$,

$$h(p_i)^2 \geq 2L_i - \log L_i - 4 \geq 2L_* - \log L_* - 4 \geq L_*,$$

because $2L - \log L$ increases for $L \geq 10$ and $L - \log L - 4 > 0$ there. Therefore $U(p) \geq (2L_* - \log L_* - 4)Q(p) \geq L_* Q(p)$. Divide Theorem 4 and its explicit staircase version by $U(p)$. $\square$

Corollary 10 (quadratic-mass concentration of near-maximizers). Let $p_{\max} \leq e^{-10}$ and $0 < \eta < 1$. For any feasible Gaussian partition with squared-centroid objective at least $U(p) - 2Q(p)$, write $b_i = \int_{A_i} x d\gamma_d(x)$. Define the deficient indices

$$I_\eta = \left\{ i : \|b_i\|^2 \leq (1 - \eta)\varphi(t_{p_i})^2 \right\}.$$

Then

$$\boxed{\sum_{i \in I_\eta} p_i^2 \leq \frac{2}{\eta L_*} \sum_i p_i^2.} \quad (27)$$

This applies in particular to the explicit staircase construction; it also applies to any attaining global maximizer.

Proof. Every one-cell deficit $\delta_i = \varphi(t_{p_i})^2 - \|b_i\|^2$ is nonnegative by (14), and the total $\sum_i \delta_i \leq 2Q(p)$ by hypothesis. For every $i \in I_\eta$,

$$\delta_i \geq \eta \varphi(t_{p_i})^2 = \eta p_i^2 h(p_i)^2 \geq \eta L_* p_i^2$$

by Corollary 9's pointwise estimate. Summation proves (27). $\square$

Dimension requirement: a rigorous obstruction and a logarithmic-dimensional example

The dimensional hypothesis $d \geq k - 1$ in the all-mass collision bound must not be silently discarded. There is a general complementary estimate for **every** ambient dimension $d \geq 1$:

$$\boxed{\mathcal{M}_d(p) \leq d \max_i p_i.} \quad (27a)$$

Indeed, writing $b_i = \int_{A_i} x d\gamma_d$, vector-valued Cauchy–Schwarz gives

$$\|b_i\|^2 \leq p_i \int_{A_i} \|x\|^2 d\gamma_d(x).$$

Summation and $\mathbb{E}\|G\|^2 = d$ give (27a).

For equal masses $p_i = 1/k$, the ceiling is $\mathcal{M}_d(p) \leq d/k$, whereas the individual-cell halfspace envelope obeys $U(p) \sim (2 \log k)/k$. Consequently, if $d = o(\log k)$,

$$\frac{\mathcal{M}_d(1/k, \dots, 1/k)}{U(p)} \rightarrow 0.$$

Thus no universal additive $C \sum p_i^2 = C/k$ envelope of the type in Theorem 4 can hold in dimensions $d = o(\log k)$: its gap in units of $\sum p_i^2$ must diverge.

For comparison, when $k = 2^d$, partition $\mathbb{R}^d$ into its 2^d orthants. Each orthant has mass $1/k$, and its Gaussian first-moment coordinates equal $\pm \sqrt{2/\pi}/k$. Therefore

$$\boxed{\frac{2d}{\pi k} \leq \mathcal{M}_d(1/k, \dots, 1/k) \leq \frac{d}{k}, \quad k = 2^d.} \quad (27b)$$

The lower bound is attained by the orthant construction; no claim of optimality in fixed dimension is made. This shows that logarithmic Gaussian dimension already supports a constant fraction of the high-dimensional optimal scale, even though a $1 - o(1)$ approximation at the minimal dimension remains a separate problem.

6. Positive-correlation Gaussian noise-stability approximation

There is also a uniform consequence for the full Gaussian noise-stability functional, not just its first derivative at zero correlation.

Let $G, G' \in \mathbb{R}^d$ be standard Gaussians with $\text{Cov}(G, G') = \rho I_d$, where $0 \leq \rho < 1$. For any partition $\mathcal{A} = (A_i)$, set

$$\text{Stab}_\rho(\mathcal{A}) = \sum_{i=1}^k \Pr\{G \in A_i, G' \in A_i\}, \quad \mathcal{N}_{d,\rho}(p) = \sup_{\gamma_d(A_i)=p_i} \text{Stab}_\rho(\mathcal{A}).$$

Theorem 11 (constructive noise-stability envelope). For $d \geq k - 1$, every positive probability vector p , and every $0 \leq \rho < 1$,

$$\boxed{Q(p) + \rho[U(p) - 2Q(p)] \leq \mathcal{N}_{d,\rho}(p) \leq Q(p) + \rho U(p) + \rho^2[1 - Q(p)]}. \quad (28)$$

More strongly, if $\mathcal{A}_{\text{stair}}$ is the explicit partition (5), then

$$\boxed{0 \leq \mathcal{N}_{d,\rho}(p) - \text{Stab}_\rho(\mathcal{A}_{\text{stair}}) \leq 2\rho Q(p) + \rho^2[1 - Q(p)]}. \quad (29)$$

The estimate is global over all measurable mass-constrained Gaussian partitions; it makes no assertion that the staircase partition is exactly optimal at any fixed $\rho > 0$.

Proof. Let H_α be the orthonormal multivariate Hermite polynomial basis of $L^2(\gamma_d)$. For the cell indicators $f_i = \mathbf{1}_{A_i}$, define

$$W_j(\mathcal{A}) = \sum_i \sum_{|\alpha|=j} |\langle f_i, H_\alpha \rangle|^2 \geq 0.$$

Mehler's identity follows by comparing coefficients in the Gaussian generating function

$$\mathbb{E} \left[e^{\langle s, G \rangle - \|s\|^2/2} e^{\langle t, G' \rangle - \|t\|^2/2} \right] = e^{\rho \langle s, t \rangle}.$$

Together with Parseval it yields, for $0 \leq \rho < 1$,

$$\text{Stab}_\rho(\mathcal{A}) = \sum_{j=0}^{\infty} \rho^j W_j(\mathcal{A}), \quad \sum_{j=0}^{\infty} W_j(\mathcal{A}) = 1.$$

The constant and first-degree coefficients are

$$W_0 = \sum_i p_i^2 = Q(p), \quad W_1 = \sum_i \left\| \int_{A_i} x d\gamma_d(x) \right\|^2.$$

Since all higher-degree coefficients are nonnegative, and $0 \leq \rho^j \leq \rho^2$ for $j \geq 2$,

$$Q + \rho W_1 \leq \text{Stab}_\rho(\mathcal{A}) \leq Q + \rho W_1 + \rho^2(1 - Q).$$

Theorem 4 gives $W_1 \leq U$ for every partition and $W_1(\mathcal{A}_{\text{stair}}) \geq U - 2Q$. Combining these inequalities proves both (28) and (29). $\square$

Remark. This theorem concerns nonnegative noise correlation. For $\rho < 0$, Hermite terms of odd order alternate in sign, so the monotonic remainder argument does not apply.

7. Scope, provenance, and what remains open

Theorem 4's additive constant **2** is a **proved universal constant**, not claimed sharp. The exact smallest possible constant in a comparison of $\mathcal{M}_d(p)$ to $U(p)$ is not determined here.

The proof does not establish which partitions optimize $\mathcal{M}_d(p)$ for any fixed $k \geq 4$; in particular it does not settle the equal-mass Standard Simplex problem. The dimension assumption $d \geq k - 1$ is essential to the stated constructive argument, although the upper bound holds for all d .

The Gaussian halfspace rearrangement bound and the normal Mills inequalities are classical. The connection to OpenAI Mathematics result 096 and our prior fixed-mass Gaussian research is recorded in README.md and LITERATURE.md. We claim the explicitly proved inequalities and construction, **not** worldwide mathematical priority before comprehensive historical search and independent peer review.

Computer verification is supplementary. The mathematical proofs of Theorems 4 and 6 and their corollaries use no solver and no numerical computation. The companion exact checker certifies finite rational mass identities and entropy inequalities for a battery of highly nonuniform and boundary-sensitive examples; it must not be mistaken for proof of the universal claims.